

VIPIN SUDHAKAR

vipinsudhakar.github.io

Software developer · B.Tech, Artificial Intelligence and Data Science (2025–2029)

vipinsudhakar007@gmail.com · +91 73059 43540 · linkedin.com/in/vipin-sudhakar · github.com/vipinsudhakar

[EDUCATION]

B.Tech in Artificial Intelligence and Data Science

2025–2029 (expected)

Amrita Vishwa Vidyapeetham, Coimbatore · CGPA 9.1/10

[PROJECTS]

TrialBridge · Hacktoberfest Hack Day Coimbatore, Team Latent

2026

- Matches cancer patients in India to recruiting clinical trials: Gemma 4 reads photos of their medical reports into a structured, evidence-quoted profile, then checks the patient against each eligibility rule of each plausible trial.
- My part: application architecture and AI integration, covering the Next.js structure, the Gemma integration, medical-document extraction, prompt development and API error handling.
- Explainable by design: code splits each trial's rules and decides the outcome; the model answers one narrow question per rule. Cut one trial check from about 3 minutes to 26 seconds by setting thinking to minimal and moving the reasoning hints into the prompt.

STACK Next.js 16, React 19, TypeScript, Gemma 4 (Gemini API), Zod, ClinicalTrials.gov API, Vercel

LIVE trialbridge-beta.vercel.app

CAAC: Content-Anchored Adaptive Chunking · Advanced DSA course project, team of four

2026

- Extends a 2026 tamper-evident logging paper (Yağız, Horasan and Yurttakal) with content-anchored chunking over a forest of Merkle trees, so a change to the log stays local.
- At 100,000 entries, absorbing an insertion takes about 62× fewer SHA-256 operations than the paper's pipeline (about 67× for an edit); proofs stay $O(\log n)$ and tamper detection stays 100%.
- Built the whole system: a Java 21 engine with RFC 6962 hashing and counted rebuild costs (367 tests), a Spring Boot + PostgreSQL API, and a React visualiser.

STACK Java 21, Spring Boot, PostgreSQL, React, TypeScript, Docker, Render

LIVE caac-cicn.onrender.com REPO github.com/vipinsudhakar/merkle-log-integrity

Filament · Personal project

2026

- Real-time multi-species Physarum (slime mould) simulation: hundreds of thousands of agents in WebGPU compute shaders (WGSL), with no 3D library in between.
- A studio to paint food, walls, repellent and lures, mix up to four species, and share any run as a link that replays bit-for-bit (fixed-point atomic deposits, a seeded PCG hash shared between TypeScript and WGSL).
- Tuned to hold 60 fps on integrated laptop GPUs.

STACK WebGPU, WGSL, TypeScript, React, Vite

LIVE vipinsudhakar.github.io/filament REPO github.com/vipinsudhakar/filament

Strand · Personal project

2026

- Rebuilt a single-file prototype as a full-stack app after finding its numbers untrustworthy (a guessed isoelectric point, a DNA molecular weight off by about 1000×): the science now runs server-side in Biopython.
- Protein and DNA analysis, plus 3D structure prediction with ESMFold proxied through the backend and rendered with 3Dmol, coloured by per-residue confidence.

STACK FastAPI, Biopython, SQLAlchemy/SQLite, React, TypeScript, Docker

LIVE strand-orgw.onrender.com REPO github.com/vipinsudhakar/Strand

[SKILLS]

Languages	TypeScript, JavaScript, Python, Java, SQL, WGSL, HTML, CSS
Frontend & graphics	React, Next.js, Astro, Vite, Tailwind CSS, GSAP, WebGPU, 3Dmol.js
Backend & data	Spring Boot, JPA / Hibernate, OpenAPI, FastAPI, Pydantic, SQLAlchemy, Node.js, PostgreSQL, Zod
AI & data science	Gemma 4, Gemini API, PyTorch, scikit-learn, XGBoost, OpenCV, NumPy, pandas, Biopython
Tools & deployment	Git, GitHub Actions, Docker, Vercel, Render, Playwright, Vitest, JUnit, pytest

[MILESTONES]

Hacktoberfest Hack Day Coimbatore · Built TrialBridge with Team Latent

2026

CAAC course project · Advanced DSA, team of four, reviewed

2026